

$$f(x) = \frac{x}{x^2-1}$$

$$D_f = \mathbb{R} \setminus \{-1; 1\}$$

$$\text{Zeros: } f(x)=0 \Leftrightarrow x=0$$

$$\lim_{x \rightarrow \pm\infty} f(x) = \lim_{x \rightarrow \pm\infty} \frac{x}{x^2-1} = \frac{\infty}{\infty} = \lim_{x \rightarrow \pm\infty} \frac{x}{x(x-\frac{1}{x})} = \lim_{x \rightarrow \pm\infty} \frac{1}{x-\frac{1}{x}} = \frac{1}{\infty} = 0$$

$$f'(x) = \left(\frac{x}{x^2-1} \right)' = \frac{(x)'(x^2-1) - (x^2-1)'x}{(x^2-1)^2} = \frac{x^2-1-2x^2}{(x^2-1)^2} = -\frac{x^2+1}{(x^2-1)^2}$$

$$\text{Zeros: } f'(x) = 0 \Leftrightarrow \text{Não tem zeros}$$

$$\begin{aligned} f''(x) &= \left(\frac{-x^2-1}{(x^2-1)^2} \right)' = \frac{(-x^2-1)'(x^2-1)^2 - ((x^2-1)^2)'(-x^2-1)}{(x^2-1)^4} = \frac{-2x(x^2-1)^2 - 2(x^2-1)(-x^2-1)}{(x^2-1)^4} = \frac{-2x(x^2-1) - 2(x^2-1)(-x^2-1)}{(x^2-1)^3} \\ &= \frac{-2x(x^2-1) - 4x(-x^2-1)}{(x^2-1)^3} = \frac{-2x^3+2x+4x^3+4x}{(x^2-1)^3} = \frac{2x(x^2+3)}{(x^2-1)^3} \end{aligned}$$

$$\text{Zeros: } f''(x) = 0 \Leftrightarrow x=0$$

x	$-\infty$		-1		0		1		$+\infty$
f(x)	0	-	~	+	0	-	~	+	0
f'(x)		- ↘	~	- ↘		- ↘	~	- ↘	
f''(x)		- ∪	~	+ ∩	0	- ∪	~	+ ∩	
		↘	~	↘	Z P.I.	↘	~	↘	

